8.1 When Solutions Aren't Unique

WORKSHEET

The question

STEP III 1994, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q4

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Take the square root of both sides: $y' = \pm 2\sqrt{y}$. Try $y = (x-c)^2$. Does it satisfy the equation? Is there an even simpler solution?

STAGE 1 Solutions through a point

Ask yourself Which parabolas $y = (x-c)^2$ pass through (a, b^2)?

STAGE 3 The whole family, and a common tangent

Pause and try Sketch several of the parabolas $y = (x-c)^2$ for different c. Which line touches every one of them? Is that line itself a solution?

STAGE 4 Why it matters

Ask yourself Start at the origin. Which solutions pass through it? Can you build a solution that follows the x-axis for a while and then leaves it?

Try this next

1. **Two rates.** (From STEP II, 2018.) Using $v = \sqrt{y}$, solve $\dfrac{\mathrm{d}y}{\mathrm{d}t} = \alpha y^{1/2} - \beta y$ (with $\alpha, \beta > 0$) and find the non-constant solution y_1 with $y_1(0) = 0$. Do the same for $\dfrac{\mathrm{d}y}{\mathrm{d}t} = \alpha y^{2/3} - \beta y$, using $v = y^{1/3}$. Why does $y = 0$ also satisfy both equations with $y(0) = 0$?

2. **The leaking bucket.** Solve $h' = -k\sqrt{h}$ with $h(0) = h_0 > 0$. When is the bucket empty? Write down two different solutions with $h(T) = 0$ for some T and $h(t) = 0$ for all $t \geq T$.

3. **Uniqueness restored.** Show that $y' = 2y$ with $y(0) = 0$ has only the solution $y = 0$. (*Hint:* consider $\frac{\mathrm{d}}{\mathrm{d}x}(ye^{-2x})$.) What property does $2y$ have that $2\sqrt{y}$ lacks?

4. **Cubes.** Find all solutions of $y' = 3y^{2/3}$ through the origin.

Further reading

- **Morris Hirsch, Stephen Smale & Robert Devaney, *Differential Equations, Dynamical Systems, and an Introduction to Chaos*** (Academic Press). Existence and uniqueness, and much of the rest of this chapter.
- **George Simmons, *Differential Equations with Applications and Historical Notes*** (CRC Press). A classic, full of history; discusses singular solutions and envelopes.
- **John Norton, "The dome: an unexpectedly simple failure of determinism"**, *Philosophy of Science* 75 (2008). Short and readable.

The full guided solution, and what the question leads to at university, is in Section 8.1:
https://beyondstep.keithmcnulty.org/ch08/non-uniqueness.html